

Palindromic Covariance Determinants and Reciprocal Thresholds in Two-Mode Linear Ornstein–Uhlenbeck Systems

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Abstract

Let a stable linear Ornstein–Uhlenbeck system have a symmetric drift whose restriction to a two-dimensional invariant subspace has relaxation rates $D_1, D_2 > 0$. We start from an exact identity for the stationary covariance Σ solving the Lyapunov equation, valid for an *arbitrary* symmetric forcing Q on that subspace:

$$4D_1D_2 \det \Sigma = \det Q + \nu^2 Q_{12}^2, \quad \nu = \frac{D_1 - D_2}{D_1 + D_2},$$

where Q_{12} is the off-diagonal entry in the drift eigenbasis. This is the 2×2 Hadamard-product determinant expansion behind Oppenheim’s inequality, applied to the Cauchy matrix $1/(D_i + D_j)$, and we claim nothing for it; it is recorded because it makes what follows transparent. No rank-one, isotropy or positivity hypothesis is needed. Applied to the affine family $Q(\tau) = q_b I + (\tau - 1)H$ with H symmetric of any rank, this gives $4D_1D_2 \det \Sigma(\tau) = q_b^2 + q_b(\text{tr } H)(\tau - 1) + K(\tau - 1)^2$ with $K = \det H + \nu^2 H_{12}^2$, and hence our main structural result: for $q_b > 0$ the determinant is a *palindromic* quadratic in τ *if and only if the deformation is trace-normalized*, $\text{tr } H = q_b$. Palindromicity then forces the reciprocal coordinate $w = \tau + 1/\tau$: the normalized determinant is exactly affine in w , yielding an exact generalized-variance threshold. Rank-one heterogeneous excitation $H = q_b e e^\top$ is the special case $\text{tr } H = q_b$ holds automatically, and recovers earlier two-node results with $\tau = r^2$. We separate the algebraic theorem from the covariance-admissible family: H positive semidefinite with $\text{tr } H = q_b$ keeps $K \geq 0$ and $Q(\tau) \succ 0$ for every $\tau > 0$. Graph-Laplacian corollaries give exact optimal thresholds 34, $35 + 15\sqrt{5}$ and $247/2$ for K_2 , P_3 and K_3 . A three-mode excitation with three distinct rates makes the determinant genuinely cubic, with leading coefficient proportional to $\prod_{i < j} (D_i - D_j)^2$, so the quadratic closure is tied to having two distinct active relaxation rates. The identity, the trace criterion, the reciprocal reduction and the admissibility bounds are machine-checked in Lean 4 with mathlib.

1 Introduction

Linear Ornstein–Uhlenbeck (OU) systems are among the few multivariate stochastic models whose stationary statistics are available in closed form: for a Hurwitz drift the stationary covariance solves an algebraic Lyapunov equation [28, 13, 29]. This makes them a natural setting in which to ask how *heterogeneous* forcing interacts with coupling, and the question has been studied directly: Barucca analyses localization in covariance matrices of coupled heterogeneous OU processes [4]; Ferreira, Metz and Barucca develop a random-matrix ensemble for OU covariances with heterogeneous temperatures

[11]; Godrèche and Luck give exact nonequilibrium stationary states for linear Langevin systems at two temperatures [14]; and the Brownian-gyrator literature studies the same two-temperature structure in a planar model [12, 10].

A scalar summary of such a covariance is its determinant, Wilks’s generalized variance [31], which is monotone in the differential entropy of the stationary Gaussian [7]. Earlier work of the author [9] analysed two diffusively coupled OU processes with unequal noise amplitudes and found that this determinant can exceed its uncoupled value at finite coupling if and only if the noise-amplitude ratio exceeds $3 + 2\sqrt{2}$ — an exact algebraic threshold in a two-node model.

The present paper grew out of trying to say exactly what makes that threshold work, and the answer turned out to be neither the graph nor the geometry we first guessed at. An earlier formulation of ours isolated a two-dimensional invariant subspace of the drift carrying a *rank-one* deformation of an isotropic forcing, and explained the resulting reciprocal structure by a self-duality of that rank-one family. That explanation is correct but not sharp.

What is actually true is simpler and more general. For a symmetric drift restricted to a two-dimensional invariant subspace, the stationary covariance obeys the exact identity

$$4D_1D_2 \det \Sigma = \det Q + \nu^2 Q_{12}^2, \quad \nu = \frac{D_1 - D_2}{D_1 + D_2}, \quad (1)$$

for *any* symmetric forcing Q — no rank-one, no isotropy, no positivity. Feeding an affine family $Q(\tau) = q_b I + (\tau - 1)H$ into (1) makes the determinant a quadratic in τ whose leading and constant coefficients coincide precisely when $\text{tr } H = q_b$. So the mechanism is a *trace normalization* of the deformation, and rank-one excitation is simply the case in which that normalization is automatic, because a unit rank-one $q_b e e^\top$ has trace q_b whatever direction e points. Isotropy of the baseline is not needed either.

The trace condition has a structural reading that we give alongside the computation. The complementary forcing $H^* = q_b I - H$ satisfies $\tau Q_H(1/\tau) = Q_{H^*}(\tau)$ unconditionally, and $\text{tr } H^* = 2q_b - \text{tr } H$; so the trace-normalized deformations are exactly the class *closed* under this involution, and on that class the coefficient K is preserved. The duality $e \mapsto e^\perp$ familiar from the rank-one model is this involution restricted to rank one. Trace balance and self-duality are therefore not two mechanisms for this family but two descriptions of one.

Only once palindromicity is established does the reciprocal coordinate appear. It is not posited as a convenient substitution: a palindromic quadratic divided by τ is an affine function of $w = \tau + 1/\tau$, and that is what produces an exact threshold. The chain is

$$\begin{aligned} \text{Lyapunov identity} &\rightarrow \text{affine forcing family} \rightarrow \text{trace normalization} \\ &\rightarrow \text{palindromic determinant} \rightarrow w = \tau + 1/\tau \rightarrow \text{exact threshold}, \end{aligned}$$

and each arrow is proved before the next is used.

We are careful throughout to separate the algebra from the modelling. Identity (1) and the trace criterion hold for any symmetric H and any real τ ; but $Q(\tau)$ is a forcing covariance only where it is positive definite, and the threshold statement needs $K \geq 0$. Both are guaranteed by taking H positive semidefinite with $\text{tr } H = q_b$, and we say so rather than quietly assuming it.

What is standard and what is not. Almost every ingredient is classical, and we are explicit about it: the OU stationary covariance and the Lyapunov equation [13]; the modal solution $\Sigma_{ij} = Q_{ij}/(D_i + D_j)$, standard in the heterogeneous-OU literature [4, 14], and its Cauchy–Hadamard form [22, 33]; the determinant identity (1) itself, which is the 2×2 Hadamard expansion behind Oppenheim’s inequality (Remark 6); the reciprocal coordinate, an invertible function of the usual

two-temperature combination (Remark 20); the reduction of a self-reciprocal polynomial through $t + 1/t$ [30]; the trace/determinant shape statistic and its relation to Mauchly's sphericity test [19, 20]; and the Newton-identity obstruction in dimension $n \geq 3$.

What we did not locate in the literature we searched is the *trace criterion* of Theorem 11, any published statement that a covariance determinant is palindromic in a forcing parameter, or the resulting exact generalized-variance threshold minimized over a coupling strength. We state that conservatively in Section 14 and claim nothing stronger.

Scope. All results are exact real algebra about a linear-Gaussian model. Nothing here is asserted about nonlinear, biological, cognitive or social systems, and no physical universality is attributed to any of the constants that appear.

2 Setting and notation

Consider a linear stochastic differential equation on $\mathbb{R}^n$,

$$dX = MX dt + B dW, \quad Q := BB^\top, \quad (2)$$

with M Hurwitz. The stationary covariance Σ is the unique solution of the algebraic Lyapunov equation

$$M\Sigma + \Sigma M^\top = -Q. \quad (3)$$

Existence and uniqueness of the stationary Gaussian law for (2) is classical [13, 29]; we take (3) as the object of study and prove statements about it.

Definition 1 (Active plane and excitation direction). **We assume throughout that M is symmetric**, so that it is orthogonally diagonalizable and the modal basis below is orthonormal; this hypothesis is load-bearing and is not a convenience. Everything in Sections 3–9 rests on the modal solution (7), which is false for non-normal drift.

Let $\Pi \subset \mathbb{R}^n$ be a two-dimensional subspace invariant under M , with eigenmodes u, v and relaxation rates $D_1, D_2 > 0$; that is, in the orthonormal basis (u, v) of Π ,

$$M|_\Pi = \text{diag}(-D_1, -D_2).$$

Let $e = cu + sv$ with $c^2 + s^2 = 1$ be a unit vector in Π , the *excitation direction*, and let $e^\perp = -su + cv$. We call Π the *active plane*. Throughout, e denotes this direction and never Euler's number.

Definition 2 (Rank-one heterogeneous forcing). For a baseline intensity $q_b > 0$ and a parameter $\tau > 0$, the forcing on Π is

$$Q(\tau) = q_b(I + (\tau - 1)ee^\top), \quad (4)$$

whose eigenvalues on Π are $q_b\tau$ along e and q_b along $e^\perp$.

Remark 3 (τ is a variance ratio). This point is a frequent source of confusion and we fix it now. The parameter τ in (4) is a ratio of *forcing eigenvalues*, i.e. of noise *variances* (or temperatures), not of noise *amplitudes*. When the two-node model of [9] is recovered in Section 12.1, its amplitude ratio r satisfies

$$\tau = r^2, \quad \text{hence} \quad \tau + \frac{1}{\tau} = \left(r + \frac{1}{r}\right)^2 - 2. \quad (5)$$

So $w = \tau + 1/\tau$ and $s(r) = r + 1/r$ are *different* reciprocal invariants of the same physical situation, related by (5); conflating them misreports a threshold by a square. We always write w for the variance invariant. This remark fixes notation only: the coordinate w plays no role until Section 5, where it is *forced* by the palindromic identity rather than chosen for convenience.

Active plane: drift eigenbasis and rank-one excitation

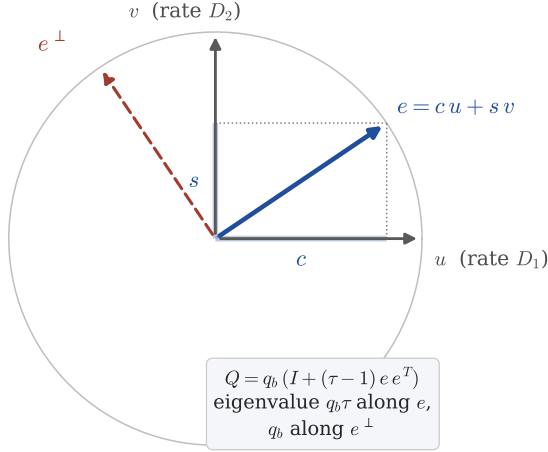


Figure 1: The active plane. The drift is diagonal in the eigenbasis (u, v) with rates D_1, D_2 ; the forcing is isotropic except for a rank-one deformation of relative size τ along the excitation direction $e = cu + sv$. The mixing c^2s^2 and the rate contrast ν are the only features of this picture that enter the determinant.

We write ν for the rate contrast and $\mathcal{A}$ for the coefficient that will govern everything:

$$\nu = \frac{D_1 - D_2}{D_1 + D_2}, \quad \mathcal{A} = c^2s^2\nu^2. \quad (6)$$

Figure 1 shows the configuration.

3 The general determinant identity

Theorem 4 (Stationary covariance on the plane). *Let $D_1, D_2 > 0$ and let $Q = \begin{pmatrix} q_{11} & q_{12} \\ q_{12} & q_{22} \end{pmatrix}$ be any symmetric forcing on Π . Then (3) restricted to Π has the unique solution*

$$\Sigma_{ij} = \frac{Q_{ij}}{D_i + D_j}, \quad i.e. \quad \Sigma = \begin{pmatrix} \frac{q_{11}}{2D_1} & \frac{q_{12}}{D_1 + D_2} \\ \frac{q_{12}}{D_1 + D_2} & \frac{q_{22}}{2D_2} \end{pmatrix}. \quad (7)$$

Proof. In the eigenbasis (3) reads entrywise $-(D_i + D_j)\Sigma_{ij} = -Q_{ij}$, and $D_i + D_j > 0$. $\square$

Equation (7) is the standard modal form of the Lyapunov solution for a symmetric drift, used throughout the heterogeneous-OU literature [4, 14, 11]; we claim no novelty for it. The next identity is the one everything rests on.

Theorem 5 (General two-mode determinant identity). *Let $D_1, D_2 > 0$ and let Q be any symmetric matrix on Π . With Σ as in (7),*

$$4D_1D_2 \det \Sigma = \det Q + \nu^2 Q_{12}^2, \quad \nu = \frac{D_1 - D_2}{D_1 + D_2}. \quad (8)$$

Proof. From (7), $\det \Sigma = \frac{q_{11}q_{22}}{4D_1D_2} - \frac{q_{12}^2}{(D_1 + D_2)^2}$, so

$$4D_1D_2 \det \Sigma = q_{11}q_{22} - \frac{4D_1D_2}{(D_1 + D_2)^2} q_{12}^2 = q_{11}q_{22} - (1 - \nu^2)q_{12}^2 = \det Q + \nu^2 q_{12}^2,$$

using $(D_1 + D_2)^2 - (D_1 - D_2)^2 = 4D_1D_2$, i.e. $4D_1D_2/(D_1 + D_2)^2 = 1 - \nu^2$. $\square$

The content of (8) is a mismatch of denominators. The diagonal entries of Σ carry $2D_i$ and the off-diagonal carries $D_1 + D_2$; if the off-diagonal carried the geometric mean $2\sqrt{D_1D_2}$ instead, the determinant would be exactly $\det Q/(4D_1D_2)$ and nothing further would happen. The arithmetic mean is larger, the deficit is precisely ν^2 , and the whole of this paper is what that deficit does.

Remark 6 ((8) is a known expansion, and we do not claim it). Theorem 5 should be read as bookkeeping, not as a discovery, and it is worth being explicit about where it sits. Writing $C_{ij} = 1/(D_i + D_j)$ for the Cauchy matrix of the rate pairs, (7) says exactly

$$\Sigma = C \circ Q \tag{9}$$

with $\circ$ the Hadamard (entrywise) product — a standard way of writing the solution of a Sylvester or Lyapunov equation with diagonal coefficients [22, 33]. For 2×2 matrices the determinant of a Hadamard product expands as

$$\det(C \circ Q) = \det C \cdot q_{11}q_{22} + C_{12}^2 \det Q, \tag{10}$$

and (8) is (10) multiplied by $4D_1D_2$, using $4D_1D_2 \det C = \nu^2$ and $4D_1D_2 C_{12}^2 = 1 - \nu^2$. Equivalently,

$$4D_1D_2 \det \Sigma = \nu^2 q_{11}q_{22} + (1 - \nu^2) \det Q, \tag{11}$$

a convex combination of $q_{11}q_{22}$ and $\det Q$ with weight ν^2 .

Equation (10) is the 2×2 case underlying Oppenheim's inequality $\det(A \circ B) \geq \det A \prod_i b_{ii}$ for positive semidefinite A, B [16, Thm. 7.8.16, eq. (7.8.17), p. 509], originally due to Oppenheim [21]. Oppenheim gives the inequality; (8) simply records the remainder, which at $n = 2$ is exactly $C_{12}^2 \det Q$. The combination “Cauchy–Hadamard solution, then Oppenheim, then a determinant bound” has been used before — [23, §4.5] does precisely this for a discrete-time steady-state covariance, stopping at the inequality. We therefore claim nothing for Theorem 5 beyond convenience of form. What we do claim begins in Section 4.

Corollary 7 (Correlation transfer). *With $\rho_\Sigma = \Sigma_{12}/\sqrt{\Sigma_{11}\Sigma_{22}}$ and $\rho_Q = q_{12}/\sqrt{q_{11}q_{22}}$,*

$$\rho_\Sigma = \frac{2\sqrt{D_1D_2}}{D_1 + D_2} \rho_Q = \sqrt{1 - \nu^2} \rho_Q. \tag{12}$$

The stationary correlation is the forcing correlation damped by the ratio of the geometric to the arithmetic mean of the relaxation rates.

Proof. Direct from (7); and $(2\sqrt{D_1D_2}/(D_1 + D_2))^2 = 4D_1D_2/(D_1 + D_2)^2 = 1 - \nu^2$. $\square$

Corollary 7 is an equivalent packaging of Theorem 5 and is likewise not claimed as new; it is the kind of statement that may well sit unreferenced in the bivariate-OU or correlation-transfer literature, and we have not exhausted that search.

Remark 8 (What is *not* assumed). Theorem 5 assumes only that Q is symmetric. It is not assumed rank one, not isotropic, not positive semidefinite, and τ has not yet been introduced. The drift, by contrast, *is* assumed symmetric on Π (Definition 1); that hypothesis is load-bearing, since (7) fails for non-normal drift.

4 Affine forcing and the trace criterion

We now deform the forcing along a one-parameter family.

Definition 9 (Affine forcing family). For a baseline intensity $q_b > 0$, a symmetric $H = \begin{pmatrix} h_1 & h_{12} \\ h_{12} & h_2 \end{pmatrix}$ and $\tau \in \mathbb{R}$, set

$$Q(\tau) = q_b I + (\tau - 1)H. \quad (13)$$

Theorem 10 (Affine determinant formula). *With $\Sigma(\tau)$ the stationary covariance of (13),*

$$4D_1 D_2 \det \Sigma(\tau) = q_b^2 + q_b(\operatorname{tr} H)(\tau - 1) + K(\tau - 1)^2, \quad K := \det H + \nu^2 H_{12}^2. \quad (14)$$

Proof. $Q(\tau)$ has off-diagonal $(\tau - 1)h_{12}$ and $\det Q(\tau) = q_b^2 + q_b(\operatorname{tr} H)(\tau - 1) + (\det H)(\tau - 1)^2$, the standard expansion of the determinant of $q_b I + (\tau - 1)H$ in dimension two. Substituting both into (8) and collecting the $(\tau - 1)^2$ terms gives (14). $\square$

Expanding (14) in powers of τ ,

$$4D_1 D_2 \det \Sigma(\tau) = \underbrace{K}_{\text{leading}} \tau^2 + (q_b \operatorname{tr} H - 2K)\tau + \underbrace{(q_b^2 - q_b \operatorname{tr} H + K)}_{\text{constant}}. \quad (15)$$

Theorem 11 (Trace criterion for palindromicity). *The leading and constant coefficients in (15) are equal if and only if $q_b(\operatorname{tr} H - q_b) = 0$. In particular, for $q_b > 0$,*

$$4D_1 D_2 \det \Sigma(\tau) \text{ is palindromic in } \tau \iff \operatorname{tr} H = q_b. \quad (16)$$

Proof. Leading minus constant equals $K - (q_b^2 - q_b \operatorname{tr} H + K) = q_b \operatorname{tr} H - q_b^2 = q_b(\operatorname{tr} H - q_b)$. For $q_b > 0$ this vanishes iff $\operatorname{tr} H = q_b$. $\square$

This is the structural statement of the paper, and it is worth saying plainly what it replaces. The reciprocal structure is *not* caused by the deformation being rank one, nor by the baseline being isotropic. It is caused by the deformation carrying exactly the baseline's share of trace. Rank one is the case where that happens for free.

There is also a structural reason for the criterion, which we give now because it explains the theorem better than the coefficient computation does — though the computation above remains the proof.

Definition 12 (Complementary forcing). For a symmetric H put $H^* := q_b I - H$.

Proposition 13 (Complementary-forcing involution). *For every symmetric H , every q_b and every $\tau \neq 0$,*

$$\tau Q_H\left(\frac{1}{\tau}\right) = Q_{H^*}(\tau), \quad (17)$$

and $(H^)^* = H$. Moreover $\operatorname{tr} H^* = 2q_b - \operatorname{tr} H$, so the trace-normalized class $\{H : \operatorname{tr} H = q_b\}$ is exactly the class closed under $H \mapsto H^*$. On that class*

$$\det H^* = \det H, \quad H_{12}^* = -H_{12}, \quad \text{hence} \quad K(H^*) = K(H). \quad (18)$$

The unique fixed point is $H = \frac{q_b}{2}I$.

Proof. $\tau Q_H(1/\tau) = \tau q_b I + (1 - \tau)H = q_b I + (\tau - 1)(q_b I - H) = Q_{H^*}(\tau)$; note that no hypothesis on $\operatorname{tr} H$ is used. Involutivity and the trace transformation are immediate. For the rest, $\det(q_b I - H) = q_b^2 - q_b \operatorname{tr} H + \det H$, which equals $\det H$ exactly when $\operatorname{tr} H = q_b$; the off-diagonal of $q_b I - H$ is $-H_{12}$, so H_{12}^2 is unchanged and $K = \det H + \nu^2 H_{12}^2$ is invariant. Finally $H^* = H$ forces $H = \frac{q_b}{2}I$. $\square$

So the identity (17) holds *unconditionally*; what the trace condition buys is that H^* lands back in the same family, with the same K . Palindromicity follows at once: $\tau^2 \det \Sigma_H(1/\tau)$ is the determinant built from $Q_{H^*}(\tau)$, and on the trace-normalized class that has the same coefficient K , hence the same polynomial.

Remark 14 (Closure is not fixedness). Closure under the involution is much weaker than $H = H^*$. The trace-normalized class is a two-parameter family; the involution acts on it with a single fixed point $\frac{q_b}{2}I$, and every other member is paired with a distinct partner. It is the *pairing*, not any individual symmetry of H , that forces the palindromic coefficients.

Remark 15 (Why the criterion is a trace, and why it is a 2×2 statement). Three features of Theorem 11 are worth isolating.

(i) *Where the coefficients come from.* Writing $\Sigma(\tau) = S_0 + \tau S_1$ with $S_1 = C \circ H$ and $S_0 = C \circ (q_b I - H)$, the leading and constant coefficients of $\det \Sigma(\tau)$ are exactly $\det(C \circ H)$ and $\det(C \circ (q_b I - H))$. So palindromicity is the single scalar equation $\det(C \circ H) = \det(C \circ (q_b I - H))$.

(ii) *Why the drift drops out.* In that equation the off-diagonal of H enters only through h_{12}^2 , which is invariant under $H \mapsto q_b I - H$ (it sends $h_{12} \mapsto -h_{12}$). Every trace of the Cauchy structure — and with it ν , and the whole content of Theorem 5 — therefore cancels, leaving only the diagonal part $q_b^2 - q_b(h_1 + h_2)$. This is why the criterion $\text{tr } H = q_b$ contains no D_1, D_2 : it is *independent of the drift*, which is not obvious from the statement.

(iii) *Why it is special to two modes.* For n modes, self-reciprocity of a degree- n determinant requires $n - 1$ coefficient matchings, not one. Already at $n = 3$ two conditions appear and $\text{tr } H = q_b$ is only one of them. The clean iff is a two-mode statement, and Section 11 is where that boundary is made quantitative.

Remark 16 (A degenerate case the criterion does not see). “Palindromic” in Theorem 11 means *leading coefficient equals constant coefficient*, which is the form used throughout. When $K = 0$ both are zero and the criterion is satisfied vacuously by a polynomial that is only of degree one: for instance $H = \text{diag}(q_b, 0)$ has $\text{tr } H = q_b$ and gives $4D_1 D_2 \det \Sigma(\tau) = q_b^2 \tau$. That is the case of an excitation aligned with a single drift eigenmode, i.e. $\mathcal{A} = 0$ in Proposition 26, and it carries no threshold (Remark 33). A reader who prefers to define palindromicity through the actual degree should read Theorem 11 together with $K \neq 0$.

Theorem 17 (Trace-normalized determinant). *If $\text{tr } H = q_b$ then*

$$4D_1 D_2 \det \Sigma(\tau) = K(\tau - 1)^2 + q_b^2 \tau = K\tau^2 + (q_b^2 - 2K)\tau + K. \quad (19)$$

Proof. Put $\text{tr } H = q_b$ in (14): the linear term becomes $q_b^2(\tau - 1)$, and $q_b^2 + q_b^2(\tau - 1) = q_b^2 \tau$. $\square$

5 The reciprocal reduction

Only now, with palindromicity in hand, does the reciprocal coordinate enter. A palindromic polynomial of even degree $2m$ in t is a degree- m polynomial in $t + 1/t$ after division by t^m ; the general statement is classical (see [30, §2]), and we use only $m = 1$, proved in line.

Definition 18. For $\tau > 0$ set $w = \tau + \frac{1}{\tau}$.

Lemma 19. *For $\tau > 0$, $w - 2 = (\tau - 1)^2/\tau \geq 0$, with equality iff $\tau = 1$. Hence $w \geq 2$ always.*

Remark 20 (w is not a new variable). If τ is read as a ratio of temperatures T_1/T_2 , then $w - 2 = (T_1 - T_2)^2/(T_1 T_2)$, a combination that appears throughout the two-temperature linear literature

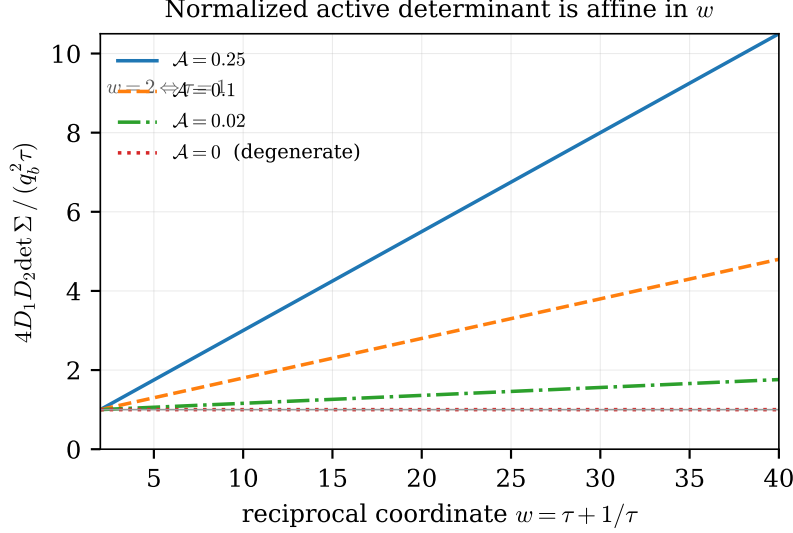


Figure 2: The normalized determinant (20) as a function of the reciprocal coordinate w , for several values of the slope. The dependence is exactly affine, with value 1 at $w = 2$ (the homogeneous point $\tau = 1$). In the rank-one case the slope is $\mathcal{A} = c^2 s^2 \nu^2 \in [0, 1/4)$; in general it is K/q_b^2 , nonnegative for an admissible family (Definition 24).

[12, 10, 14]. So w is an invertible function of a standard quantity and we claim nothing for the coordinate itself. What the results below assert is that the normalized determinant is *exactly affine* in it, and precisely when.

Corollary 21 (Reciprocal reduction). *If $\text{tr } H = q_b$ and $\tau > 0$, then*

$$\frac{4D_1 D_2 \det \Sigma(\tau)}{\tau} = q_b^2 + K(w - 2), \quad \text{equivalently} \quad \frac{4D_1 D_2 \det \Sigma(\tau)}{q_b^2 \tau} = 1 + \frac{K}{q_b^2}(w - 2). \quad (20)$$

Proof. Divide (19) by τ and use Lemma 19: $K(\tau - 1)^2/\tau + q_b^2 = K(w - 2) + q_b^2$. $\square$

Figure 2 plots (20). The dependence on the heterogeneity is exactly affine in w , with slope K/q_b^2 and value 1 at $w = 2$.

6 Admissibility: algebra versus covariance

Everything above is an identity between polynomials and holds for every symmetric H and every real τ . Two further things are needed before it describes a stochastic system, and we state them separately rather than folding them into the hypotheses.

Proposition 22 (Sign of K). *$K = \det H + \nu^2 H_{12}^2 = h_1 h_2 - (1 - \nu^2) h_{12}^2$. If H is positive semidefinite then $K \geq 0$. If H is indefinite, K may be negative: $H = \text{diag}(2, -1)$ has $\text{tr } H = 1$ and $K = -2$. Moreover $K = 0$ is attained, e.g. $H = \text{diag}(q_b, 0)$.*

Proof. $0 \leq 1 - \nu^2 \leq 1$ by Proposition 28 below, so $K \geq \det H$; if H is positive semidefinite then $\det H \geq 0$ and $\nu^2 H_{12}^2 \geq 0$. The two examples are direct. $\square$

Proposition 23 (Positivity of the forcing). *Let H be positive semidefinite with $\text{tr } H = q_b > 0$. Then $Q(\tau) \succ 0$ for every $\tau > 0$.*

Proof. H has eigenvalues η and $q_b - \eta$ with $0 \leq \eta \leq q_b$. The eigenvalues of $Q(\tau) = q_b I + (\tau - 1)H$ are $q_b + (\tau - 1)\eta_i$, which for $\tau > 0$ are bounded below by $\min\{q_b, \tau q_b\} > 0$. Equivalently, by (14) with $\text{tr } H = q_b$, $\det Q(\tau) = q_b^2 \tau + (\tau - 1)^2 \det H > 0$ and $\text{tr } Q(\tau) = q_b(1 + \tau) > 0$. $\square$

Definition 24 (Admissible family). We call (13) *admissible* when $q_b > 0$ and H is positive semidefinite with $\text{tr } H = q_b$. By Propositions 22 and 23 an admissible family has $K \geq 0$ and $Q(\tau) \succ 0$ for all $\tau > 0$, so (20) is a statement about genuine forcing covariances and its slope is nonnegative.

7 Rank-one excitation as a special case

Corollary 25 (Rank-one heterogeneous excitation). *Let $e = cu + sv$ with $c^2 + s^2 = 1$ and put $H = q_b ee^\top$, i.e. $Q(\tau) = q_b(I + (\tau - 1)ee^\top)$. Then $\text{tr } H = q_b$ automatically, H is positive semidefinite, $\det H = 0$, $H_{12} = q_b cs$, and*

$$K = q_b^2 c^2 s^2 \nu^2, \quad \frac{4D_1 D_2 \det \Sigma(\tau)}{q_b^2 \tau} = 1 + \mathcal{A}(w - 2), \quad \mathcal{A} := c^2 s^2 \nu^2. \quad (21)$$

Proof. $\text{tr}(q_b ee^\top) = q_b \|e\|^2 = q_b$ and $\det(q_b ee^\top) = 0$ since $ee^\top$ has rank one. Then $K = 0 + \nu^2 (q_b cs)^2$. Divide by q_b^2 in (20). $\square$

This is the theorem of our earlier formulation, now with its hypotheses explained: the rank-one model works because a unit rank-one deformation is trace-normalized *for every direction* e , and it is admissible because such an H is positive semidefinite. The direction enters only through $c^2 s^2$.

Proposition 26 (Degeneracy and size, rank-one case). *For the rank-one family, $\mathcal{A} \geq 0$ and*

$$\mathcal{A} = 0 \iff c = 0 \text{ or } s = 0 \text{ or } D_1 = D_2, \quad (22)$$

while the reciprocal contribution $\mathcal{A}(w - 2)$ vanishes iff $\mathcal{A} = 0$ or $\tau = 1$.

The readings are: $c = 0$ or $s = 0$, the excitation lies in a single drift eigenmode, so forcing and drift are simultaneously diagonal and the covariance cannot mix them; $D_1 = D_2$, no relaxation contrast; $\tau = 1$, no heterogeneity. A nonzero reciprocal contribution needs *both* anisotropies, in the drift and in the forcing.

Remark 27. Equation (22) is specific to rank one. For general trace-normalized H the coefficient is K , and K can be nonzero with $\nu = 0$: the isotropic scaling $H = \frac{q_b}{2} I$ gives $K = q_b^2/4$ for any rates. So the degeneracy analysis must be read as a statement about the rank-one family, not about (16).

Proposition 28 (Modal-mixture bound). *If $c^2 + s^2 = 1$ then $c^2 s^2 \leq \frac{1}{4}$, with equality iff $c^2 = s^2 = \frac{1}{2}$. If $D_1, D_2 > 0$ then $\nu^2 < 1$. Hence $0 \leq \mathcal{A} < \frac{1}{4}$ in the rank-one case.*

Proof. $\frac{1}{4} - c^2 s^2 = (c^2 - \frac{1}{2})^2 \geq 0$; and $(D_1 + D_2)^2 - (D_1 - D_2)^2 = 4D_1 D_2 > 0$. $\square$

8 The rank-one duality is a specialization

The rank-one isotropic model has a duality that makes the reciprocal structure visually obvious, and it is often how the phenomenon is first noticed. It is worth recording precisely because it is *not* an independent mechanism: it is Proposition 13 restricted to rank one.

Proposition 29 (Rank-one complement). *Let $c^2 + s^2 = 1$ and $e^\perp = -su + cv$. Then for $H = q_b ee^\top$,*

$$H^* = q_b(I - ee^\top) = q_b e^\perp e^{\perp\top}, \quad (23)$$

and consequently (17) reads

$$Q_e(\tau) = \tau Q_{e^\perp} \left(\frac{1}{\tau} \right). \quad (24)$$

Proof. $ee^\top + e^\perp e^{\perp\top} = I$ for an orthonormal pair, so $q_b I - q_b ee^\top = q_b e^\perp e^{\perp\top}$. Substituting into (17) gives (24). $\square$

In other words, “the excitation direction flips to its orthogonal complement” and “the deformation is replaced by its complement $q_b I - H$ ” are the same statement, the first being the rank-one case of the second. Likewise “trace balance” and “self-duality” are not two mechanisms for this family: by Proposition 13 the trace condition *is* the condition that the family is closed under the duality. A unit rank-one deformation satisfies it automatically for every direction e , which is why the rank-one model looks self-dual without any extra hypothesis.

Remark 30 (The duality picture is specific to an isotropic baseline). What *is* genuinely weaker than palindromicity is the existence of any such duality at all, once the baseline is allowed to be anisotropic. Take $Q_0 = \text{diag}(p, q)$ with $p \neq q$ in the drift eigenbasis and deform along e with the scale $\mu = 1/(e^\top Q_0^{-1} e)$, i.e. $Q(\tau) = Q_0 + \mu(\tau - 1)ee^\top$. Then:

- $\det Q(\tau) = \tau \det Q_0$ exactly — that normalization is precisely what makes τ the *determinant ratio* of the forcing;
- consequently $\det Q(\tau)$ is linear in τ while $Q_{12}(\tau) = \mu(\tau - 1)cs$, so by (8) the determinant is $\tau \det Q_0 + \nu^2 \mu^2 c^2 s^2 (\tau - 1)^2$, manifestly palindromic;
- but (24) fails for $p \neq q$, and Proposition 13 does not apply because the baseline is not $q_b I$.

So outside the isotropic-baseline family, palindromicity can survive with no forcing-level duality available. The clean equivalence of Proposition 13 is a statement about the affine family (13), not about determinant palindromicity in general.

Remark 31 (Non-commuting baselines). For a baseline with a nonzero off-diagonal in the drift eigenbasis, the analogous condition still has solutions, but the required scale then depends on the drift as well as on the forcing, which we do not regard as a natural parameterization. We record only that commutation of the baseline with the drift is sufficient, not necessary, and do not pursue the exceptional family here.

9 The threshold theorem

Theorem 32 (Exact threshold). *Let $\Phi > 0$ and $C > 0$ and put $V = \Phi(1 + C(w - 2))$. Then*

$$V > 1 \iff w > w^*, \quad w^* = 2 + \frac{1 - \Phi}{\Phi C}. \quad (25)$$

Proof. $w - (2 + \frac{1-\Phi}{\Phi C}) = (\Phi(1 + C(w - 2)) - 1)/(\Phi C)$ identically, and $\Phi C > 0$. $\square$

In the applications $C = K/q_b^2$ by Corollary 21, and $C = \mathcal{A}$ in the rank-one case.

Remark 33 (Edge cases, stated rather than hidden). Since $w \geq 2$ always (Lemma 19), three regimes must be distinguished.

1. $\Phi < 1$: then $w^* > 2$ and (25) is a genuine *heterogeneity threshold*. This is the case in all graph applications below.
2. $\Phi = 1$: $w^* = 2$, so $V > 1$ for every $\tau \neq 1$. Any heterogeneity at all suffices.
3. $\Phi > 1$: $w^* < 2$ is unreachable and $V > 1$ for every admissible w — *automatic enhancement*, with no threshold.
4. $C = 0$: excluded by the hypothesis $C > 0$. Then $V = \Phi$ identically and no threshold exists; this is the degenerate case of Proposition 26, and also the case $K = 0$ of Proposition 22.

Finally, w^* is a threshold in the *variance* invariant; converting to an amplitude ratio requires (5).

10 Passive modes and the ambient system

Theorem 11 concerns a 2×2 block. We now record, as an analytic statement, how that block sits inside an n -dimensional system — and we are careful about what is proved versus what is formalized.

Proposition 34 (Determinant factorization). *Let M be symmetric and Hurwitz on $\mathbb{R}^n$ with eigenmodes $\psi_1, \dots, \psi_n$ and rates $D_1, \dots, D_n > 0$, so that $\Pi = \text{span}(\psi_1, \psi_2)$ and its orthogonal complement are both M -invariant. Suppose the forcing is*

$$Q = q_b(I + (\tau - 1)ee^\top) \quad (26)$$

with e a unit vector whose modal support lies in $\Pi = \text{span}(\psi_1, \psi_2)$, i.e. $e = c\psi_1 + s\psi_2$ with $c^2 + s^2 = 1$. Then Σ block-diagonalizes as $\Sigma|_\Pi \oplus \Sigma|_{\Pi^\perp}$ and

$$\det \Sigma = \underbrace{\frac{q_b^2 \tau [1 + \mathcal{A}(w - 2)]}{4D_1 D_2}}_{\text{active block}} \cdot \prod_{j \geq 3} \frac{q_b}{2D_j} = \frac{q_b^n \tau [1 + \mathcal{A}(w - 2)]}{2^n \prod_{j=1}^n D_j}. \quad (27)$$

Proof. In the modal basis Q has entries $\tilde{Q}_{ij} = q_b \delta_{ij} + q_b(\tau - 1)\tilde{e}_i \tilde{e}_j$ with $\tilde{e} = (c, s, 0, \dots, 0)$. Hence $\tilde{Q}_{ij} = 0$ whenever $i \leq 2 < j$ or $j \leq 2 < i$, so $\tilde{Q}$ is block diagonal. By (7), $\tilde{\Sigma}_{ij} = \tilde{Q}_{ij}/(D_i + D_j)$ inherits the same block structure, and the determinant factors. The active factor is Theorem 21; each passive mode contributes $\tilde{\Sigma}_{jj} = q_b/(2D_j)$. $\square$

Remark 35 (The hypotheses are all needed). Four separate conditions are doing work here, and forcing support alone is not enough: (i) M symmetric, so an orthonormal modal basis exists and (7) applies; (ii) Π invariant under M ; (iii) $\Pi^\perp$ invariant, automatic for symmetric M ; (iv) the *baseline* block-diagonal with respect to $\Pi \oplus \Pi^\perp$ — automatic here because the baseline is the isotropic $q_b I$, but false in general; and (v) the deformation supported in Π . If the baseline couples the two blocks, or if the excitation leaks into a passive mode, Σ does not block-diagonalize and the determinant does not factor.

Remark 36 (Formalized versus documented). Proposition 34 is proved here analytically. The machine-checked theorem (Section 16) is the *explicit active block* — Theorems 4, 11, 21 and 32. The general invariant-subspace reduction of Proposition 34 is documented analytically and is *not* formalized. We are explicit about this because the distinction is easy to blur: no claim is made that Lean verifies an arbitrary $n \times n$ invariant-subspace reduction.

10.1 Three dimensions that must not be conflated

Proposition 34 invites a piece of vocabulary, and the audit of this paper showed that a single loose phrase will not do. Three different integers are in play, and the theorems depend on the third:

<i>ambient dimension</i>	n , the size of the system
<i>modal support dimension</i>	the number of eigenmodes on which the deformation acts
<i>distinct-rate dimension</i>	the number of <i>distinct</i> relaxation rates among those modes

The hypothesis of Theorems 5–11 is that the **distinct-rate dimension is two**. The ambient dimension is irrelevant; the stationary law has full support on $\mathbb{R}^n$ and every mode is excited, the passive ones at the baseline intensity. The modal support dimension may exceed two provided the extra modes share a rate — which is exactly what happens for K_3 in Section 12.3, and is why that example is not a counterexample.

We therefore say the configuration has *distinct-rate dimension two* and avoid saying that the system “is really two-dimensional”. This is interpretation attached to a hypothesis, not a separate theorem, but it is the interpretation that makes the failure boundary in Section 11 sayable.

11 The failure boundary: three distinct rates

The quadratic closure is exactly a two-distinct-rate phenomenon, and we can say so with a formula rather than a dimension count.

Proposition 37 (Cubic response). *Let M be symmetric with rates $D_1, D_2, D_3 > 0$ on a three-dimensional invariant subspace, and let $Q(\tau) = q_b(I + (\tau - 1)ee^\top)$ with $e = \sum_i c_i \psi_i$, $\sum_i c_i^2 = 1$. Then $\det \Sigma(\tau)$ is a cubic in τ whose leading coefficient is*

$$\frac{q_b^3 c_1^2 c_2^2 c_3^2 (D_1 - D_2)^2 (D_1 - D_3)^2 (D_2 - D_3)^2}{\prod_{i < j} (D_i + D_j)^2} \Big/ \prod_i 2D_i, \quad (28)$$

up to the normalization $\prod_i 2D_i$ used to clear denominators. It vanishes if and only if some $c_i = 0$ — the support drops to two modes — or some $D_i = D_j$.

Two consequences. First, (28) is a clean generalization of $\mathcal{A} = c^2 s^2 \nu^2$: the same product structure, one factor of $(D_i - D_j)^2$ and one of c_i^2 per mode. Second, the correct statement of the failure boundary is *generic*, not absolute:

A genuinely three-mode excitation with three *distinct* relaxation rates generically produces a cubic response, and no single reciprocal coordinate can absorb it. If a rate is repeated, or a modal weight vanishes, the cubic term disappears and the quadratic closure returns.

We have not investigated whether the cubic can be palindromic on some measure-zero set, and we do not claim it cannot.

12 Graph corollaries

Remark 38 (Weak coupling). Before specializing, one general fact puts the minimization below in context. If $\Sigma(\kappa)$ solves $(aI + \kappa S)\Sigma + \Sigma(aI + \kappa S) = Q$ for symmetric S and fixed positive-definite Q , then $\frac{d}{d\kappa} \log \det \Sigma|_{\kappa=0} = -\text{tr } S/a$, independently of Q : differentiating gives $\Sigma' = -(S\Sigma_0 + \Sigma_0 S)/(2a)$

with $\Sigma_0 = Q/(2a)$, whence $\text{tr}(\Sigma_0^{-1}\Sigma') = -\text{tr} S/a$. For a graph Laplacian $\text{tr} L = 2|E|$, so the initial rate is $-2|E|/a < 0$: infinitesimal coupling always contracts the generalized variance, and any enhancement is necessarily a finite-coupling phenomenon. This generalizes the two-node statement $A'(0) < 0$ of [9].

We now specialize to a concrete family. Let G be a connected graph on n nodes with Laplacian L , and take

$$M = -(aI + \kappa L), \quad a > 0, \kappa \geq 0, \quad (29)$$

so the modal rates are $D_j = a + \kappa \lambda_j$ for Laplacian eigenvalues $0 = \lambda_0 \leq \lambda_1 \leq \dots \leq \lambda_{n-1}$. Let the forcing be (26) with e a node indicator: heterogeneity at a single node. Define

$$\Phi(\kappa) = \prod_{j \geq 1} \frac{a}{a + \kappa \lambda_j}, \quad V(\kappa, \tau) = \frac{\det \Sigma(\kappa, \tau)}{\det \Sigma(0, \tau)}. \quad (30)$$

Corollary 39 (Graph form of the threshold). *Suppose the node indicator e has modal support in exactly two eigenmodes, with rates D_1, D_2 and weights c^2, s^2 . Then*

$$V(\kappa, \tau) = \Phi(\kappa) [1 + \mathcal{A}(\kappa) (w - 2)], \quad \mathcal{A}(\kappa) = c^2 s^2 \nu(\kappa)^2, \quad (31)$$

and $V > 1$ exactly when $w > w^*(\kappa) = 2 + (1 - \Phi)/(\Phi \mathcal{A})$.

Proof. By (27), $\det \Sigma(\kappa, \tau) = q_b^n \tau [1 + \mathcal{A}(w - 2)] / (2^n \prod_j D_j)$. At $\kappa = 0$ all rates equal a , so $\nu = 0$, $\mathcal{A} = 0$, and $\det \Sigma(0, \tau) = q_b^n \tau / (2^n a^n)$. Dividing gives (31) with $\prod_j a/D_j = \Phi$. Theorem 32 then applies. $\square$

The optimal threshold for a given graph and forced node is $w_{\min}^* = \min_{\kappa > 0} w^*(\kappa)$: the least heterogeneity that can ever produce enhancement, optimized over coupling strength. Figure 3 shows the three curves computed below.

12.1 K_2 : recovering the two-node model

For K_2 with $a = 1$, the Laplacian eigenvalues are 0 and 2, so $D_1 = 1$, $D_2 = 1 + 2\kappa$, and $\Phi = 1/(1 + 2\kappa)$. A node indicator has weights $c^2 = s^2 = \frac{1}{2}$ on the two modes, so $c^2 s^2 = \frac{1}{4}$ and

$$\mathcal{A}(\kappa) = \frac{1}{4} \nu^2, \quad \nu = \frac{1 - (1 + 2\kappa)}{1 + (1 + 2\kappa)} = \frac{-\kappa}{1 + \kappa}.$$

Corollary 39 gives

$$V(\kappa, \tau) = \frac{1}{1 + 2\kappa} \left[1 + \frac{1}{4} \frac{\kappa^2}{(1 + \kappa)^2} (w - 2) \right], \quad (32)$$

which is exactly the volume ratio of [9] after the substitution $\tau = r^2$. Minimizing $w^*(\kappa)$ over $\kappa > 0$ gives

$$\kappa^* = 1, \quad w_{\min}^* = 34. \quad (33)$$

By (5), $w = 34$ corresponds to $r + 1/r = 6$, i.e. $r = 3 + 2\sqrt{2}$, which is exactly the threshold of [9]. The coupling $\kappa^* = 1$ is the tangency at which the enhancement window of that paper closes to a point.

Remark 40. This is the sense in which the two-node result is a corollary rather than a separate theorem. It is also a check on Remark 3: the constant 34 is a value of $w = \tau + 1/\tau$, while 6 is a value of $r + 1/r$, and $34 = 6^2 - 2$.

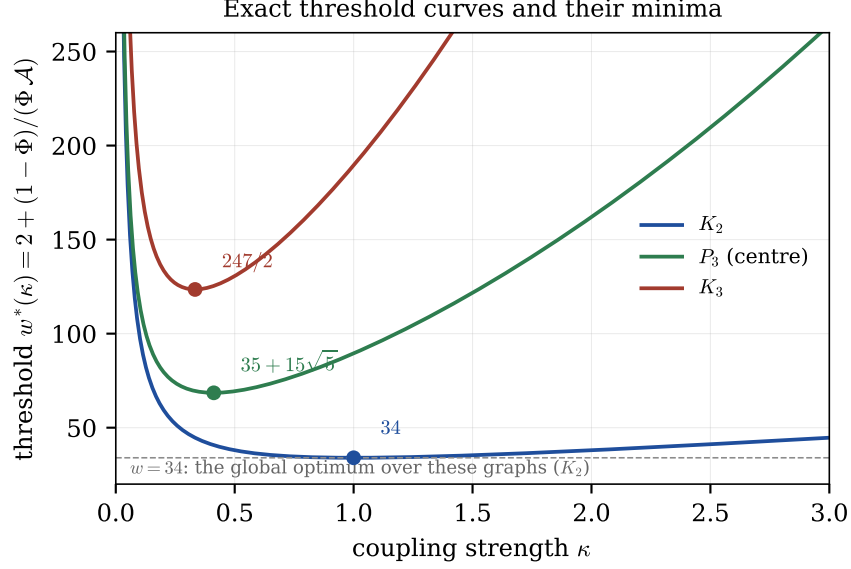


Figure 3: Exact threshold curves $w^*(\kappa)$ for K_2 , P_3 (centre node forced) and K_3 , with their exact minima marked. Each curve is the rational function derived in Sections 12.1–12.3; the marked points are the exact algebraic constants, not numerical optima.

12.2 P_3 : the path on three nodes

For P_3 the Laplacian has spectrum $\{0, 1, 3\}$ with orthonormal eigenvectors

$$\psi_0 = \frac{1}{\sqrt{3}}(1, 1, 1), \quad \psi_1 = \frac{1}{\sqrt{2}}(1, 0, -1), \quad \psi_2 = \frac{1}{\sqrt{6}}(1, -2, 1).$$

Force the *centre* node, $e = (0, 1, 0)$. Its modal components are $\langle \psi_0, e \rangle = 1/\sqrt{3}$, $\langle \psi_1, e \rangle = 0$, $\langle \psi_2, e \rangle = -2/\sqrt{6}$. The vanishing middle component is the crucial structural fact: *the centre-node excitation has no support on the $\lambda = 1$ mode*, so its active support is exactly two-dimensional and Proposition 34 applies with

$$c^2 = \frac{1}{3} (\lambda = 0), \quad s^2 = \frac{2}{3} (\lambda = 3), \quad D_1 = 1, \quad D_2 = 1 + 3\kappa,$$

the $\lambda = 1$ mode being passive. Then $\nu = -3\kappa/(2 + 3\kappa)$ and

$$\mathcal{A}(\kappa) = \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{9\kappa^2}{(2 + 3\kappa)^2} = \frac{2\kappa^2}{(2 + 3\kappa)^2}, \quad \Phi(\kappa) = \frac{1}{(1 + \kappa)(1 + 3\kappa)}.$$

Substituting into $w^* = 2 + (1 - \Phi)/(\Phi \mathcal{A})$ and simplifying gives the exact curve

$$w_{P_3}^*(\kappa) = \frac{27\kappa^3 + 72\kappa^2 + 64\kappa + 16}{2\kappa}. \quad (34)$$

Remark 41 (The two-mode hypothesis is not vacuous: P_3 forced at an end). It matters that the centre node was chosen. Forcing an *end* node of P_3 , $e = (1, 0, 0)$, gives modal components $(1/\sqrt{3}, -1/\sqrt{2}, 1/\sqrt{6})$ — all three nonzero — so the modal support dimension is three and the three rates $1, 1 + \kappa, 1 + 3\kappa$ are distinct. By Proposition 37 the determinant is then a genuine cubic in τ , with leading coefficient $\kappa^6/[4(\kappa + 2)^2(2\kappa + 1)^2(3\kappa + 2)^2]$, and it is *not* palindromic. The affine form (31) fails outright for that configuration. So the hypothesis of Corollary 39 is a real restriction, and the simplest asymmetric case already violates it.

Corollary 42 (P_3 constant). $\min_{\kappa>0} w_{P_3}^*(\kappa) = 35 + 15\sqrt{5}$, attained at $\kappa^* = \frac{\sqrt{5}-1}{3}$.

Proof. The exact factorization

$$27\kappa^3 + 72\kappa^2 + 64\kappa + 16 - (35 + 15\sqrt{5}) \cdot 2\kappa = 9\left(\kappa - \frac{\sqrt{5}-1}{3}\right)^2 (3\kappa + 6 + 2\sqrt{5})$$

holds identically. For $\kappa > 0$ the second factor is positive, so the left side is nonnegative, giving $w_{P_3}^*(\kappa) \geq 35 + 15\sqrt{5}$; the double root shows equality at $\kappa^* = (\sqrt{5}-1)/3$. $\square$

12.3 K_3 : the triangle

For K_3 the spectrum is $\{0, 3, 3\}$. A node indicator has $c^2 = 1/3$ on ψ_0 and the remaining weight $2/3$ inside the *degenerate* $\lambda = 3$ eigenspace. Because that eigenspace is two-dimensional with a single common rate, one may rotate within it so that one basis vector carries the entire projection; the excitation then has support in exactly two modes and Proposition 34 applies with $c^2 = 1/3$, $s^2 = 2/3$, $D_1 = 1$, $D_2 = 1 + 3\kappa$. The third mode is passive with the same rate $1 + 3\kappa$. Hence $\mathcal{A}$ is as for P_3 while Φ differs:

$$\mathcal{A}(\kappa) = \frac{2\kappa^2}{(2 + 3\kappa)^2}, \quad \Phi(\kappa) = \frac{1}{(1 + 3\kappa)^2},$$

giving the exact curve

$$w_{K_3}^*(\kappa) = \frac{81\kappa^3 + 162\kappa^2 + 112\kappa + 24}{2\kappa}. \quad (35)$$

Corollary 43 (K_3 constant). $\min_{\kappa>0} w_{K_3}^*(\kappa) = \frac{247}{2}$, attained at $\kappa^* = \frac{1}{3}$.

Proof. $81\kappa^3 + 162\kappa^2 + 112\kappa + 24 - \frac{247}{2} \cdot 2\kappa = 81\kappa^3 + 162\kappa^2 - 135\kappa + 24 = 3(3\kappa - 1)^2(3\kappa + 8)$, which is nonnegative for $\kappa > 0$ and vanishes at $\kappa = 1/3$. $\square$

Remark 44 (Why K_3 is not a counterexample). K_3 is the instructive case for Section 10.1. A node indicator in K_3 has *modal support dimension three*: it meets ψ_0 and both vectors of the $\lambda = 3$ eigenspace. Nevertheless its *distinct-rate dimension is two*, because that eigenspace is degenerate — both of its modes relax at $a + 3\kappa$. Within a degenerate eigenspace any orthonormal basis is admissible, so one may rotate to align a single basis vector with the projection, after which the hypothesis of Proposition 34 holds with $c^2 = 1/3$, $s^2 = 2/3$. Consistently with Proposition 37, the cubic coefficient carries a factor $(D_i - D_j)^2$ that vanishes for the repeated rate, so the response is quadratic and palindromic after all.

Where the relevant rates are *distinct*, as for P_3 , no such freedom exists and the vanishing of a modal component is a genuine constraint on the graph and the forced node. This is precisely the distinction that Section 10.1 is designed to keep visible.

12.4 Summary of the corollaries

Table 1 collects the results. Numerically $35 + 15\sqrt{5} \approx 68.54$ and $247/2 = 123.5$, both larger than the K_2 value 34: among these three configurations, the two-node system is the one for which the least heterogeneity suffices. We deliberately stop at three worked cases rather than cataloguing families of graphs; see Section 15 for what we do *not* claim about general graphs.

Table 1: The three worked graph corollaries. All constants are exact and derived from Corollary 39; none is fitted. “Formalized” refers to the Lean development of Section 16.

Model	Active rates	c^2, s^2	κ^*	$w_{\min}^*$	Formalized
K_2	1, $1 + 2\kappa$	$\frac{1}{2}, \frac{1}{2}$	1	34	yes
P_3 (centre)	1, $1 + 3\kappa$	$\frac{1}{3}, \frac{2}{3}$	$\frac{\sqrt{5}-1}{3}$	$35 + 15\sqrt{5}$	yes
K_3 (any node)	1, $1 + 3\kappa$	$\frac{1}{3}, \frac{2}{3}$	$\frac{1}{3}$	$\frac{247}{2}$	yes

13 Two-dimensional spectral closure

A separate two-dimensional phenomenon explains why an earlier spectral argument closed with a single scalar. For a positive-definite 2×2 matrix with eigenvalues $\mu_1 \geq \mu_2 > 0$, write $q = \mu_1/\mu_2 \geq 1$ and

$$J_2(\Sigma) = \frac{(\text{tr } \Sigma)^2}{\det \Sigma}. \quad (36)$$

Proposition 45. $J_2 = q + \frac{1}{q} + 2$, and $q \mapsto q + 1/q$ is strictly increasing on $[1, \infty)$. Hence J_2 determines q uniquely.

Proof. $(\mu_1 + \mu_2)^2/(\mu_1\mu_2) = \mu_1/\mu_2 + \mu_2/\mu_1 + 2$. Strict monotonicity of $q + 1/q$ on $[1, \infty)$ is elementary. $\square$

This is standard. Scale-free trace/determinant functionals of a covariance matrix are the classical sphericity diagnostics, going back to Mauchly [19] and treated textbook-style by Muirhead [20, Ch. 8]; trace/determinant eigenvalue bounds of this type are likewise classical [32, 16]. We claim no novelty. It is recorded because it is the reason a single scale-free scalar suffices in dimension two — the same reason, in a different guise, that one reciprocal coordinate suffices in Theorem 21.

13.1 Failure in higher dimension

Proposition 46 (Non-identifiability for $n \geq 3$). *For $n = 3$ the pair $(\text{tr } \Sigma, \det \Sigma)$ does not determine the spectral shape: the positive triples $(1, 8, 12)$ and $(2, 3, 16)$ have the same trace 21 and the same determinant 96, hence the same value of every scale-free invariant built from trace and determinant alone, but different extreme-eigenvalue ratios 12 and 8.*

This too is background mathematics, not a new theorem. By Newton’s identities a spectrum is determined by all n elementary symmetric functions, of which trace and determinant are the complete list only when $n = 2$; the failure above is the elementary consequence. The invariant theory behind such statements is classical, and the statistical form is Mauchly’s [19, 20].

We stress the relation to Section 10.1. Proposition 46 is about the *spectrum* of an $n \times n$ covariance; Proposition 34 is about the *support of the heterogeneity*. They are different statements, but they point the same way: the one-coordinate closure used throughout this paper is a two-dimensional phenomenon. If the excitation genuinely spans three or more modes with distinct rates, neither the palindromic determinant nor the spectral closure is available, and we make no claim about that case.

14 Relation to existing work, and what we claim

Standard ingredients. The following are classical and are used as such: the stationary OU covariance and the Lyapunov equation [13, 29, 25]; the modal solution (7) for symmetric drift, standard in heterogeneous-OU work [4, 14, 11]; the determinant as Wilks’s generalized variance [31] and its entropy reading [7]; graph Laplacian eigenmodes and their use in network dynamics [2]; the reduction of a self-reciprocal polynomial through $t + 1/t$ [30]; the trace/determinant shape statistic [19, 20, 32, 16].

Closest located prior art. The nearest genuine antecedent is the work of Summers, Cortesi and Lygeros on submodularity and controllability in complex networks [26], which proves that log det of the controllability Gramian is a monotone increasing and submodular *set* function of the chosen input columns (see also [1]). That is the qualitative predecessor of the present result, and it differs in four ways that matter: it concerns a controllability Gramian rather than the stationary covariance solving (3); it is a set function over discrete input subsets rather than a continuous one-parameter family; the forcing is *added* with no baseline constraint, whereas (4) reshapes a fixed isotropic baseline; and it yields no closed form, no palindromic structure and no threshold. We are sharpening a qualitative monotonicity statement into an exact identity in a narrower setting, not contradicting or subsuming it.

Two-temperature linear Langevin systems [14, 12, 10] share the model class exactly — in particular [14, §2.10] gives the explicit 2×2 stationary covariance for arbitrary friction and diffusion — but their scalar diagnostics are entropy production, irreversibility and heat flux, not the covariance determinant. Likewise [5] analyses precisely the two-degree-of-freedom two-temperature geometry studied here and tracks an individual variance entry rather than the generalized variance. Low-rank structure in Lyapunov solutions is well studied [25, 27], but for solution approximation and singular-value bounds rather than determinants.

What we claim. We are deliberately narrow. The determinant identity (8) is the 2×2 Hadamard expansion (10) applied to the Cauchy matrix, i.e. the equality case of Oppenheim’s inequality with the remainder written out (Remark 6); it is not claimed. The Cauchy–Hadamard form (9) of a Sylvester solution is standard [22, 33]. The reciprocal coordinate is an invertible function of the usual two-temperature combination (Remark 20) and is not claimed. The reduction of a palindromic polynomial through $t + 1/t$ is classical [30]. The closest published use of the same pipeline we located is [23, §4.5], which forms a discrete-time steady-state covariance as a Cauchy–Hadamard product and applies Oppenheim to bound its determinant, stopping at the inequality.

Determinants of Lyapunov solutions. There is a substantial control-theory literature on the solution of the algebraic Lyapunov equation, and it is worth saying what shape it has: it consists of *bounds* — on eigenvalues, on the trace, and on the determinant — rather than of exact expressions; see the survey [18]. We did not locate an exact determinant identity in that line. Theorem 5 is exact but, as Remark 6 explains, it is exact only because $n = 2$ makes the Hadamard expansion terminate, so it does not compete with that literature.

Gaussian quantum steady states. The same Lyapunov machinery governs the steady state of two coupled quantum oscillators attached to baths at different temperatures, and there $\frac{1}{2} \log \det \sigma$ is the standard Rényi-2 entropy — so this is the literature in which an equivalent of our construction would most plausibly hide. We checked it and did not find one. The nearest setting is [6], which

solves exactly this Lyapunov problem for two oscillators with two baths, uses $\frac{1}{2} \log \det \sigma$, and plots its figure of merit against the bath-occupation ratio with an extremum at ratio one — but the quantity studied is the entropy-production rate, and no determinant threshold appears. Elsewhere in that literature the exchange symmetry $T_1 \leftrightarrow T_2$ is noted explicitly and the stronger ratio structure is *not* claimed; [17, § discussion of Fig. 5] states the invariance and then argues against a reduction to $|\beta_1 - \beta_2|$, proposing no ratio-symmetric form. Thresholds in that literature are typically in *absolute* temperature for a separability predicate, not in a temperature *ratio* for a determinant crossover.

Two structural obstructions keep the quantum case outside our hypotheses, and we record them rather than gesture at them. First, a two-mode Gaussian system carries momenta, so its covariance is 4×4 and its drift is *not* symmetric — the Hamiltonian part sees to that — so the modal solution (7) does not apply. Second, bath asymmetry is precisely what mixes the normal modes, so the 4×4 problem does not block-diagonalize into 2×2 blocks in the regime of interest.

Remark 47 (The reciprocal phenomenon is broader than this theorem). It is worth being candid that palindromicity is not peculiar to the hypotheses used here. For the canonical underdamped pair — two oscillators coupled by a spring, each damped at rate γ_i and driven by its own bath at temperature T_i — we computed the stationary 4×4 covariance directly. Its determinant is a *quartic* in (T_1, T_2) , and it is palindromic in the temperature ratio exactly when the two damping rates are equal; unequal frequencies do no harm, unequal damping destroys it. In that case $\det \sigma / (T_1 T_2)^2$ is a *quadratic* in w , not an affine function of it.

So the reciprocal structure appears there too, but in a different form and for a different reason: the condition is on the *drift* (equal damping), whereas Theorem 11 imposes a condition on the *forcing* ($\text{tr } H = q_b$) and returns an exactly affine dependence. We did not locate this quartic observation in the literature either, and we do not develop it here; it is recorded to mark the boundary of what the present theorem does and does not explain. (Verified symbolically; see the reproducibility script.)

Remark 48 (A quantum boundary case, and what transfers). One model makes the boundary unusually sharp. Hofer et al. [15] report the steady-state moments of two coherently coupled bosonic modes, each locally damped at rate κ_α into its own thermal bath, in the standard local-Lindblad treatment. They report moments; they report no determinant and no purity. From those moments one obtains, writing $m_\alpha = n_\alpha + \frac{1}{2}$ and $N_{ij} = \langle a_i^\dagger a_j \rangle$,

$$\det(N + \tfrac{1}{2}I) = \frac{4g^2 \kappa_h^2 m_h^2 + \kappa_h \kappa_c (8g^2 + (\kappa_h + \kappa_c)^2) m_h m_c + 4g^2 \kappa_c^2 m_c^2}{(4g^2 + \kappa_h \kappa_c)(\kappa_h + \kappa_c)^2}. \quad (37)$$

The coefficients of m_h^2 and m_c^2 are $4g^2 \kappa_h^2$ and $4g^2 \kappa_c^2$, so (37) is palindromic in the thermal ratio *exactly when* $\kappa_h = \kappa_c$ — which is our criterion in this model, since $Q(\tau) = \kappa \text{diag}(\tau, 1)$ has $q_b = \kappa$ and $H = \text{diag}(\kappa, 0)$, so $\text{tr } H = q_b$ holds precisely for equal damping. For $\kappa_h = \kappa_c = \kappa$ the expression collapses and

$$\frac{\det(N + \tfrac{1}{2}I)}{m_h m_c} = \frac{g^2(w + 2) + \kappa^2}{4g^2 + \kappa^2}, \quad w = \tau + \frac{1}{\tau}, \quad \tau = \frac{m_h}{m_c}, \quad (38)$$

again affine in w .

What this comparison is for is the dissociation it exhibits. The *criterion* transfers and the *reciprocal signature* transfers; the determinant identity (8) and the threshold do not. The identity does not because the steady-state equation here has a complex drift and Hermitian conjugation rather than the real-symmetric structure of (3): the modal rates are $\kappa/2 \pm ig$, so $\nu = -2ig/\kappa$ is purely imaginary and $\nu^2 < 0$. We therefore do not apply Theorem 5 to this model. The threshold does not transfer because ∂_{g^2} of (38) equals $\kappa^2(w - 2)/(\kappa^2 + 4g^2)^2 \geq 0$: the normalized determinant is monotone in the coupling, with no interior extremum of the kind that produces the classical threshold.

The moral is worth stating plainly: *palindromicity alone does not imply the threshold phenomenon*. Equation (37) is derived here from the moments reported in [15]; we attribute the steady state to those authors and the determinant to this paper, and we located no equivalent palindromic formulation in the literature reviewed.

Determinants of network covariances. One line of work does use a covariance determinant as a network performance measure: Siami and Motee’s *uncertainty volume* is a log-determinant of the steady-state output covariance of a noisy consensus network, and sits alongside the more familiar trace-type measures [24]. It differs from the present setting in every respect that matters here — the noise is homogeneous, the drift is a pure Laplacian, and the optimization is over which edges to add rather than over a scalar coupling, with no threshold in a heterogeneity parameter. The network-coherence literature it grows out of [2] is otherwise uniformly trace-type. Separately, “generalized variance” in the Granger-causality and integrated-information literature [3] denotes the determinant of a *residual* covariance, a different matrix from $\Sigma(\kappa)$; the two should not be conflated.

What we did not locate, in the literature we searched, is:

1. the *trace criterion* of Theorem 11 — a condition $\text{tr } H = q_b$ controlling a palindromic symmetry of a covariance determinant in a deformation parameter;
2. any statement that a covariance or Gramian determinant is palindromic or self-reciprocal in a forcing or temperature parameter;
3. the resulting exact generalized-variance threshold, minimized over a coupling strength.

We describe these as an *apparently distinct assembly* and make no priority claim. Should an equivalent result exist, the right reading of this paper is as an exact derivation and machine-checked formalization of it. The search was English-language web and arXiv only, with no access to MathSciNet, zbMATH or publisher full text, and no matrix-analysis textbook was consulted in full; Theorem 5 in particular is exactly the sort of statement that may appear as an exercise in a standard reference.

Relation to the author’s other work. The two-node model of [9] is recovered in Section 12.1 as the corollary $\tau = r^2$, $c^2 = s^2 = 1/2$; that paper is the motivating special case and is cited rather than reproduced. Separately, [8] records a reciprocal invariant $\text{tr}(T) \text{tr}(T^{-1})$ determining a two-dimensional shape ratio up to inversion, and is relevant as conceptual background on one-coordinate completeness in two dimensions. We state explicitly that the present theorem is derived independently from Lyapunov covariance dynamics; [8] is invariant-theoretic context, not a source for it.

15 Limitations

1. The results are exact statements about a linear-Gaussian model. No nonlinear, biological, cognitive, social or physical-universality claim is made, and no significance attaches to the numerical value of any constant beyond this model class.
2. **The drift is assumed symmetric** on the relevant invariant subspace, so that the modal basis is orthonormal and (7) holds. Non-normal drift is outside the scope of every theorem here.
3. Theorems 5–11 concern a 2×2 block, i.e. a configuration of distinct-rate dimension two (Section 10.1). With three distinct rates and genuine three-mode support the response is generically cubic (Proposition 37) and nothing here applies.

4. The palindromicity criterion $\text{tr } H = q_b$ is an identity between polynomials. It carries no positivity: for indefinite H the coefficient K may be negative and $Q(\tau)$ may leave the positive-definite cone, in which case the formula is still true but is no longer a statement about a forcing covariance. The admissible family of Definition 24 is where the stochastic reading is valid.
5. The threshold theorem additionally requires $C > 0$, i.e. $K > 0$. The case $K = 0$ is attainable (Proposition 22) and has no threshold at all.
6. Proposition 34 — the n -dimensional reduction — is proved analytically and is *not* machine-checked; and it needs all five hypotheses of Remark 35, not merely that the forcing is supported in the active plane.
7. The formalization verifies algebra, not stochastic-process theory. That the Σ of (3) is the stationary covariance of (2) is classical and assumed.
8. “Distinct-rate dimension” (Section 10.1) is interpretive vocabulary attached to a hypothesis, not an independent theorem.
9. We work out three graph corollaries. We claim no general characterization of which graphs and forced nodes satisfy the hypothesis, and no general bound on $w_{\min}^*$ over graphs.
10. Optimizing w^* over κ treats coupling as freely tunable and the forcing geometry as fixed; other optimizations give other constants.
11. The literature searches behind Section 14 used web search, the arXiv API, Crossref and OpenAlex full-text search, together with a number of primary sources read in full. They did *not* reach Google Scholar, Scopus, Web of Science, MathSciNet or zbMATH — each was attempted and returned a captcha, paywall redirect or HTTP error — and arXiv’s own search is metadata-only, so a result appearing as an unnamed mid-paper remark would be invisible to them. No matrix-analysis monograph was consulted in full; a Hadamard-product problems section is a plausible home for (10). The two-mode *quantum* Gaussian literature has since been searched as well (Section 14), using the arXiv, Crossref, OpenAlex and INSPIRE full-text APIs; non-English literature was not searched. “Not located” throughout should therefore be read as the outcome of a documented but non-exhaustive sweep, not as a claim of absence.

16 Machine-checked verification

The central algebraic content is formalized in Lean 4 with mathlib, in the public repository <https://github.com/innerlightr-wq/ou-threshold-lean>. Within it:

- the generalized results of this paper live in `OUCorridor/ActivePlaneGeneral.lean`, with the earlier rank-one development in `OUCorridor/ActivePlane.lean` and `OUCorridor/SpectralShape.lean`;
- supporting material is in `Covariance.lean`, `Threshold.lean`, `MainTheorem.lean`, `ResidualThreshold.lean` and `CrossThreshold.lean`;
- frozen snapshots of the exact sources, environment and build are kept at `verified/palindromic-generalization-2026-09/` (this paper) and `verified/active-plane-2026-09/` (the earlier record, unchanged and still verifying).

Environment. Lean 4.33.1 (x86_64-unknown-linux-gnu), with mathlib pinned at revision

0df444a360eaa60ab8c11dca51a86af692955474.

Audit. A grep of the library for `sorry`, `admit` and `axiom` returns no matches. The 28 theorems of the generalized development audited here, together with the 28 audited for the earlier snapshot, report exactly the three standard axioms of mathlib’s classical foundation under `#print axioms: propext`, `Classical.choice`, `Quot.sound`. There is no custom axiom and no `sorryAx`. The recorded build completes with 8716 jobs.

What is not formalized. As stated in Remark 36 and Table 2: the n -dimensional factorization of Proposition 34, the interpretive content of Section 10.1, and all stochastic-process theory. Lean checks exact real algebra on matrices. The modelling step — that (3) describes the stationary covariance of (2) — is classical and assumed.

17 Reproducibility

The manuscript directory `paper/active-plane-theorem/` is self-contained: it holds `main.tex`, its own copy of `references.bib`, the figures, the build script, the build log, and `manuscript_checks.py`. Every displayed identity in this paper is re-derived and verified symbolically by that script using exact arithmetic (SymPy), with no floating-point step in any claim: it solves the Lyapunov equation from scratch rather than checking a pre-written solution, extracts the determinant’s coefficients in τ and compares leading against constant, derives Φ and $\mathcal{A}$ for P_3 and K_3 from the graph Laplacians, and confirms each exact minimum by algebraic factorization. The script reports 126/126 checks passing.

Exact commit hashes, the branch name and file digests are recorded in the accompanying `README.md` and `SHA256SUMS.txt` at the state in which this draft was built.

Acknowledgments

AI-assisted tools were used for symbolic checking, literature organization and citation verification, code support, construction and checking of the Lean development, and editorial assistance. Claims identified in Table 2 as formalized were checked by Lean’s kernel under the pinned environment recorded in Section 16, which does not depend on the assistant’s correctness; the remaining claims were verified symbolically by the reproducibility script. The author reviewed the manuscript and takes responsibility for it. This draft has not undergone independent human peer review.

A Open items

Recorded so that the gaps are explicit rather than silent.

1. **Characterizing admissible configurations.** Which pairs (graph, forced node) give a node indicator with modal support in exactly two eigenmodes? The P_3 centre node works because of a symmetry-induced vanishing; K_3 works because of eigenvalue degeneracy. A general criterion is open, and is a question about graph symmetry rather than about the present theorem.
2. **Beyond rank one.** For a rank- k deformation with $k \geq 2$, or an excitation spanning three or more distinct-rate modes, the determinant is a polynomial of higher degree in the heterogeneity parameters and no single reciprocal coordinate is expected to suffice. We have not investigated whether a weaker structured reduction survives.

3. **Non-normal drift.** The modal formula (7) assumes a symmetric drift on the active plane. The non-normal case is open here.

References

- [1] Athanasios C. Antoulas. *Approximation of Large-Scale Dynamical Systems*. Society for Industrial and Applied Mathematics, January 2005.
- [2] Bassam Bamieh, Mihailo R. Jovanovic, Partha Mitra, and Stacy Patterson. Coherence in large-scale networks: Dimension-dependent limitations of local feedback. *IEEE Transactions on Automatic Control*, 57(9):2235–2249, September 2012.
- [3] Adam B. Barrett and Anil K. Seth. Practical measures of integrated information for time-series data. *PLoS Computational Biology*, 7(1):e1001052, 2011.
- [4] Paolo Barucca. Localization in covariance matrices of coupled heterogeneous ornstein-uhlenbeck processes. *Physical Review E*, 90(6), December 2014.
- [5] Nicola Boffi and Paolo De Gregorio. Variance resonance in weakly coupled harmonic oscillators driven by thermal gradients. *Entropy*, 26(12):1087, 2024.
- [6] Matteo Brunelli and Mauro Paternostro. Irreversibility and correlations in coupled quantum oscillators. arXiv:1610.01172, 2016.
- [7] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, April 2005.
- [8] Elias De Jesús. The dual diagonal and reconstruction of orthogonal hyperrectangles: An alternative invariant basis. Zenodo, April 2026.
- [9] Elias De Jesús. Heterogeneity-driven expansion and synchronization collapse in coupled ornstein-uhlenbeck systems. Zenodo, August 2026.
- [10] Victor Dotsenko, Anna Maciołek, Oleg Vasilyev, and Gleb Oshanin. Two-temperature langevin dynamics in a parabolic potential. *Physical Review E*, 87(6), June 2013.
- [11] Leonardo S. Ferreira, Fernando L. Metz, and Paolo Barucca. Random matrix ensemble for the covariance matrix of ornstein-uhlenbeck processes with heterogeneous temperatures. *Physical Review E*, 111(1), January 2025.
- [12] Roger Filliger and Peter Reimann. Brownian gyrator: A minimal heat engine on the nanoscale. *Physical Review Letters*, 99(23), December 2007.
- [13] Crispin W. Gardiner. *Handbook of Stochastic Methods*. Springer Berlin Heidelberg, 2004.
- [14] Claude Godrèche and Jean-Marc Luck. Characterising the nonequilibrium stationary states of ornstein-uhlenbeck processes. *Journal of Physics A: Mathematical and Theoretical*, 52(3):035002, December 2019.
- [15] Patrick P. Hofer, Martí Perarnau-Llobet, L. David M. Miranda, Géraldine Haack, Ralph Silva, Jonatan B. Brask, and Nicolas Brunner. Markovian master equations for quantum thermal machines: local versus global approach. *New Journal of Physics*, 19(12):123037, 2017.

- [16] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge University Press, Cambridge, 2nd edition, 2013. Oppenheim’s inequality is Theorem 7.8.16, eq. (7.8.17), p. 509. The citation key retains the year of the Crossref online record (2012); the printed second edition is dated 2013.
- [17] Jen-Tsung Hsiang and B. L. Hu. Quantum entanglement at high temperatures? Bosonic systems in nonequilibrium steady state. *Journal of High Energy Physics*, 2015(11):090, 2015.
- [18] W. H. Kwon, Y. S. Moon, and S. C. Ahn. Bounds in algebraic Riccati and Lyapunov equations: a survey and some new results. *International Journal of Control*, 64(3):377–389, 1996.
- [19] John W. Mauchly. Significance test for sphericity of a normal n -variate distribution. *The Annals of Mathematical Statistics*, 11(2):204–209, June 1940.
- [20] Robb J. Muirhead. *Aspects of Multivariate Statistical Theory*. Wiley, March 1982.
- [21] A. Oppenheim. Inequalities connected with definite Hermitian forms. *Journal of the London Mathematical Society*, s1-5(2):114–119, 1930.
- [22] Davide Palitta and Sanda Lefteriu. An efficient, memory-saving approach for the Loewner framework. *Journal of Scientific Computing*, 91, 2022.
- [23] Christian M. Schlick and Bruno Demissie. Project dynamics and emergent complexity. arXiv:1101.0754, 2011.
- [24] Milad Siami and Nader Motee. Growing linear dynamical networks endowed by spectral systemic performance measures. *IEEE Transactions on Automatic Control*, 63(7):2091–2106, 2018.
- [25] V. Simoncini. Computational methods for linear matrix equations. *SIAM Review*, 58(3):377–441, January 2016.
- [26] Tyler H. Summers, Fabrizio L. Cortesi, and John Lygeros. On submodularity and controllability in complex dynamical networks. *IEEE Transactions on Control of Network Systems*, 3(1):91–101, March 2016.
- [27] Alex Townsend and Heather Wilber. On the singular values of matrices with high displacement rank. *Linear Algebra and its Applications*, 548:19–41, July 2018.
- [28] G. E. Uhlenbeck and L. S. Ornstein. On the theory of the brownian motion. *Physical Review*, 36(5):823–841, September 1930.
- [29] P. Vatiwutipong and N. Phewchean. Alternative way to derive the distribution of the multivariate ornstein–uhlenbeck process. *Advances in Difference Equations*, 2019(1), July 2019.
- [30] Ricardo Vieira. Polynomials with symmetric zeros. In *Polynomials - Theory and Application*. IntechOpen, May 2019.
- [31] S. S. WILKS. Certain generalizations in the analysis of variance. *Biometrika*, 24(3-4):471–494, 1932.
- [32] Henry Wolkowicz and George P.H. Styan. Bounds for eigenvalues using traces. *Linear Algebra and its Applications*, 29:471–506, February 1980.
- [33] Igor Yadykin. Spectral decomposition of Gramians of continuous linear systems in the form of Hadamard products. *Mathematics*, 12(1):36, 2023.

Table 2: Status of every result. LEAN = machine-checked by Lean’s kernel; SYM = verified symbolically in exact arithmetic by `manuscript_checks.py`; AN = proved analytically in this paper only; INT = interpretation.

This paper	Lean name	Status
Theorem 4	<code>genCov_lyapunov</code> , <code>genCov_unique</code>	LEAN
Theorem 5	<code>det_genCov_eq</code>	LEAN
Theorem 10	<code>det_affineCov_eq</code> , <code>det_affineCov_expand</code>	LEAN
Theorem 11	<code>affine_palindromic_iff</code> , <code>affine_palindromic_iff_trace</code>	LEAN
Proposition 13	<code>affineForcing_complement</code> , <code>complementForcing_involutive</code> , <code>complementForcing_trace</code> , <code>complementForcing_traceNormalized</code> , <code>complementForcing_det</code> , <code>affineCoeff_complement</code>	LEAN
Remark 14	<code>complementForcing_fixed_iff</code>	LEAN
Theorem 17	<code>det_affineCov_traceNormalized</code>	LEAN
Lemma 19	<code>two_le_reciprocalExcitation</code> , <code>reciprocalExcitation_eq_two_iff</code>	LEAN
Corollary 21	<code>normalizedAffineDet_eq</code> , <code>normalizedAffineDet_div_eq</code>	LEAN
Proposition 22	<code>affineCoeff_nonneg_of_posSemidef</code>	LEAN
Proposition 23	<code>det_affineForcing_pos_of_posSemidef</code>	LEAN
Corollary 25	<code>rankOne_trace</code> , <code>affineCoeff_rankOne</code> , <code>affineForcing_rankOne</code> , <code>affineCov_rankOne</code> , <code>det_activeCov_palindromic_of_general</code>	LEAN
Proposition 26	<code>activeCoeff_eq_zero_iff</code>	LEAN
Proposition 28	<code>mixing_le_quarter</code> , <code>rateContrast_sq_lt_one</code>	LEAN
Proposition 29	<code>complementForcing_rankOne</code> , <code>activeForcing_self_dual</code>	LEAN
Theorem 32	<code>volumeRatio_gt_one_iff</code>	LEAN
Section 12.1	<code>volume_div_volume0_eq</code> , <code>reciprocalExcitation_sq</code>	LEAN
Corollary 42	<code>p3_thresholdCurve_eq</code> , <code>p3_threshold_min/_attained</code>	LEAN
Corollary 43	<code>k3_thresholdCurve_eq</code> , <code>k3_threshold_min/_attained</code>	LEAN
Proposition 45	<code>J2_eq_reciprocalInvariant</code> , <code>eigen_ratio_unique_of_J2</code>	LEAN
Proposition 46	<code>exists_same_trace_det_different_ratio</code>	LEAN
Remark 30 (anisotropic counterexample)	—	SYM
Proposition 37 (cubic response)	—	SYM
Remark 38 (weak-coupling law)	—	SYM
Proposition 34 (n -dimensional factorization)	—	AN
Section 10.1 (distinct-rate dimension)	—	INT